Spatio-Temporal Detection and Biogeochemical Profiling of *Noctiluca scintillans* Blooms in the Arabian Sea Using Multispectral Imagery and Hydrodynamic Reanalysis

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ABSTRACT

In recent years, large recurring winter blooms of the green dinoflagellate *Noctiluca scintillans* in the northern Arabian Sea have created major environmental concerns by replacing traditional diatom populations, lowering dissolved oxygen levels, and harming coastal fisheries. Detecting and tracking these blooms using satellites is challenging because standard empirical ocean color algorithms struggle to identify mixotrophic cells that contain green endosymbionts (*Pedinomonas noctilucae*). In addition, regular satellite images only show surface 2D views without accounting for deeper water mixing and weather conditions. In this paper, we present an integrated Earth observation and data processing pipeline that combines multispectral satellite imagery (MODIS-Aqua, VIIRS, and OLCI) with ocean physical data from Copernicus Marine Service (CMEMS) and meteorological reanalysis from ECMWF ERA5. We use narrow-band remote sensing reflectance (R_{rs}) ratios at 531 nm and 510 nm along with a multilayer neural network atmospheric correction (NN-V) to accurately identify *Noctiluca* cells and recover satellite images affected by high solar zenith angles. To analyze long-term patterns, we apply Cochrane-Orcutt autoregression and generalized least squares across multi-band reflectance over a 20-year period, separating genuine ecological trends from background environmental noise. This framework gives coastal managers a practical diagnostic tool that connects surface satellite observations with ocean mixing and dust deposition, supporting marine conservation goals under UN Sustainable Development Goal 14.

Keywords*—Noctiluca scintillans*, Ocean Color Remote Sensing, Remote Sensing Reflectance, Bio-Optical Inversion, Arabian Sea, Environmental Time-Series.

I. INTRODUCTION

The seasonal reversal of monsoon winds strongly controls ocean physics, chemistry, and biological productivity across the northern Indian Ocean [2], [3]. During the winter northeast monsoon (from December to March), cold dry winds cool the sea surface and cause evaporative cooling, which makes surface waters sink and drives convective mixing in the northern Arabian Sea [12]. This mixing brings essential nutrients such as dissolved nitrate and silicate from deeper layers into the sunlit surface zone. Historically, these nutrient-rich waters supported large seasonal blooms of diatoms, which served as the primary food source for copepods and commercial fish species while helping transfer organic carbon to the ocean floor [11]. Over the past twenty years, however, ship surveys and satellite observations show that the ecosystem is undergoing a major shift: natural diatom communities are increasingly being replaced by massive blooms of the green dinoflagellate *Noctiluca scintillans* [1], [15]. Unlike the red variety of *Noctiluca*, green *Noctiluca* contains millions of photosynthetic green algae endosymbionts (*Pedinomonas noctilucae*). This allows the organism to grow both by using sunlight for photosynthesis and by actively eating remaining diatoms and microscopic zooplankton [13]. Because common marine copepods avoid eating *Noctiluca* due to its large size and cellular ammonia buildup, the bloom biomass is not transferred up the food chain to fish, but instead goes to jellyfish and salps. When these massive blooms die and sink, bacterial decay consumes large amounts of oxygen, causing severe coastal hypoxia that kills fish and damages coral reefs, conflicting with UN Sustainable Development Goal 14 [14].

Detecting and monitoring these harmful algal blooms is an important challenge in environmental data science and marine informatics. Satellite visible sensors offer the only practical way to monitor such large marine regions on a regular basis [7], [19]. However, current satellite processing workflows suffer from three primary limitations. First, standard empirical ocean color algorithms, such as NASA's OCx band ratios, assume that all water optical properties depend strictly on chlorophyll-*a* concentration. Mixotrophic green *Noctiluca*, however, has an unusual pigment spectrum with a distinct reflectance peak at 531 nm and strong light absorption at 510 nm due to its endosymbionts; as a result, standard algorithms misclassify bloom pixels, underestimate cell numbers, or confuse blooms with coastal sediments [17]. Second, operational remote sensing pipelines usually treat satellite rasters as flat 2D surface images [10]. Spaceborne optical sensors can only measure light from the top surface layer ($Z_{pd} \approx 1/K_d$), meaning they cannot observe subsurface vertical velocities, mixed-layer convective deepening, or oxygen minimum zones that trigger bloom growth [9]. Third, long-term environmental assessments often rely on single-variable chlorophyll time series, which contain heavy year-to-year natural swings and strong temporal autocorrelation (red noise), meaning it can take thirty to forty years of data before a genuine climate-driven trend can be separated from background noise [5], [6], [8].

To solve these practical issues, this study develops an integrated computational methodology that connects satellite multi-band reflectance with physical oceanographic reanalysis data. By reviewing twenty foundational and recent

papers across bio-optics, satellite calibration, atmospheric correction, and time-series statistics, we create an end-to-end framework tailored for the northern Arabian Sea. Our approach applies narrow-band reflectance ratios at $R_{rs}(531)$ and $R_{rs}(510)$ to calculate *Noctiluca* cell densities [1], uses a multilayer neural network atmospheric correction to restore winter satellite images affected by high solar zenith angles [4], integrates daily Copernicus Marine Service (CMEMS) hydrodynamic mixing layers [2], and applies multivariate autoregressive regression across visible spectral bands to detect emerging decadal shifts over a 20-year satellite record [5].

II. RELATED WORKS

To build a solid methodology, we organize the literature into four connected areas: (A) Regional Arabian Sea Bloom Dynamics and Hydro-Atmospheric Drivers, (B) Bio-Optical Remote Sensing and Satellite Inversion Modeling, (C) Satellite Data Calibration and Atmospheric Correction Under Complex Geometry, and (D) Statistical Trend Detection and Climate Time of Emergence.

A. Regional Arabian Sea Bloom Dynamics and Hydro-Atmospheric Drivers

Pandi et al. [1] carried out field experiments in the northeastern Arabian Sea using water samples and optical data collected from the research vessel FORV *Sagar Sampada*. They showed that the reflectance band ratio $R_{rs}(531)/R_{rs}(510)$ has a very strong linear correlation (R^2_{adj} = 0.99, $p < 10^{-4}$) with *Noctiluca* cell counts. This happens because the green endosymbionts cause a reflectance peak at 531 nm and carotenoid absorption at 510 nm, making this ratio a reliable index to track bloom density in both coastal and open waters.

Shi and Wang [2] analyzed satellite images from the VIIRS sensor between 2012 and 2019, comparing them with earlier SeaWiFS data and GODAS ocean models. They found that summer blooms near the western Arabian Sea are driven by strong coastal upwelling and upward Ekman pumping, where vertical water velocities reach around 1 m/day at 70 m depth. On the other hand, winter blooms in the northern Arabian Sea occur because cold northeast monsoon winds cool the surface and deepen the mixed layer to about 80 m, bringing nutrients up to the surface.

Kuttippurath et al. [3] examined primary productivity and chlorophyll changes across the north Indian Ocean from 1998 to 2019 using OC-CCI satellite data, CMEMS models, and MERRA-2 dust records. They observed that primary productivity decreased by -0.048 mg m^{-3} day^{-1} per year, while sea surface temperature increased by +0.02 °C per year. They also noted that wind patterns carrying mineral dust from deserts in the Arabian Peninsula supply iron and phosphorus, which can trigger algae blooms even without strong upwelling.

Goes et al. [11] explained the broader climate factors behind the ecological shift in the Arabian Sea. They linked declining winter snow cover over the Himalayan-Tibetan Plateau to changes in monsoon winds. This reduced winter convective mixing and lowered the amount of dissolved silicate reaching the surface compared to nitrogen. Because diatoms require silicate to build their cell walls, this imbalance allowed green *Noctiluca*, which does not need silicate, to multiply rapidly.

Lakshmi et al. [12] used CTD instruments and water samples to study the conditions that trigger winter blooms. Using Sverdrup critical depth theory, they showed that bloom formation happens in two stages: winter cooling first mixes the water column deeply to pull nitrate into the surface layer, and then daytime sunlight warms the upper layer, making it shallow enough for the photosynthetic endosymbionts to get enough light to grow.

Asha Devi et al. [13] studied how *Noctiluca* blooms disrupt the local marine food chain in the northeastern Arabian Sea. Under normal conditions, diatoms are eaten by copepods, which are then eaten by fish. However, copepods avoid eating *Noctiluca* because of its large cell size and high ammonia content. As a result, the energy moves into jellyfish, salps, and smaller microzooplankton instead of commercial fish species.

Raj et al. [14] reported on the severe impact of decaying *Noctiluca* blooms along the southern coast of India. When the large bloom died, bacteria consumed huge amounts of oxygen to break down the cells, causing dissolved oxygen levels to drop below 1.5 mg/L. This sudden drop in oxygen killed local fish and damaged coral reefs, showing why early satellite detection is critical for coastal areas.

Xiang et al. [15] conducted laboratory incubation experiments off the coast of Pakistan to understand why *Noctiluca* outcompetes diatoms in winter. They observed that diatoms stop growing once dissolved nutrients like silicate run out. In contrast, green *Noctiluca* can switch between photosynthesis using its green endosymbionts and directly feeding on remaining diatom cells, allowing it to survive and grow even in nutrient-poor conditions.

Sarma et al. [16] investigated water chemistry and oxygen levels during winter blooms in the northeastern Arabian Sea. By measuring carbon and nitrogen isotopes along with dissolved oxygen, they proved that blooms start in well-oxygenated surface water (oxygen saturation above 85%) during convective mixing. Hypoxia only develops later when the dead bloom sinks and decomposes, clarifying that low oxygen is a result of the bloom rather than its cause.

Kuttippurath et al. [18] investigated how tropical storms and cyclones affect phytoplankton growth in the north Indian Ocean. Strong cyclonic winds create intense mixing that pulls cold, nutrient-rich water up from deeper layers, triggering large algal blooms in desert ocean areas. This highlights the importance of including meteorological events in ocean monitoring systems.

B. Bio-Optical Remote Sensing and Satellite Inversion Modeling

Cael et al. [5] showed that satellite trend analysis can be improved by studying full spectral reflectance rather than only chlorophyll-*a*. Analyzing a 20-year dataset (2002–2022) of MODIS-Aqua images across ten visible channels, and applying the Cochrane-Orcutt method to remove serial autocorrelation, they detected clear climate-driven trends in about 40% of the world's oceans. Their work proved that

multi-band reflectance reveals ecosystem changes decades sooner than single-variable chlorophyll records.

Dutkiewicz et al. [6] used a 3D ocean ecosystem model combined with radiative transfer calculations to predict ocean color changes over the 21st century under climate change. They showed that reflectance in the blue-green range (around 487–512 nm) has less year-to-year variation than chlorophyll-*a*, allowing earlier detection of climate signals. They also found that shifting levels of organic matter cause standard empirical chlorophyll formulas to lose accuracy over long periods.

Bisson et al. [9] validated satellite particulate backscattering (b_{bp}) products from MODIS, VIIRS, and OLCI by comparing them with 2,486 measurements taken by autonomous Biogeochemical Argo (BGC-Argo) floats at 700 nm. Testing the GSM, QAA, and GIOP models, they found that the standard GSM model overestimates backscattering by up to 50% in low-to-medium biomass waters because it assumes a fixed spectral slope (γ = 1). In contrast, QAA and GIOP calculate γ dynamically based on blue-to-green ratios, providing much higher accuracy.

Loisel et al. [20] developed the LS2 semi-analytical inversion model to calculate seawater absorption (a) and particulate backscattering (b_{bp}) from remote sensing reflectance (R_{rs}) and sun angle without making rigid assumptions about the spectral shape of water constituents. Using Hydrolight simulations and global field data, they demonstrated high retrieval accuracy ($R^2 > 0.96$) in both open ocean and coastal waters, making LS2 a reliable inversion method.

Baliarsingh et al. [17] designed an optical classification method using MODIS-Aqua satellite data to distinguish between green mixotrophic *Noctiluca* and red heterotrophic *Noctiluca* in the northern Indian Ocean. By examining absorption from prasinoxanthin and chlorophyll-*b*, they showed that green *Noctiluca* has higher reflectance in green wavelengths ($R_{rs}(531)$, $R_{rs}(551)$) and lower reflectance in blue ($R_{rs}(443)$), which helps separate it from regular diatom blooms in satellite imagery.

C. Satellite Data Calibration and Atmospheric Correction Under Complex Geometry

Li et al. [4] evaluated VIIRS satellite reflectance data from 2012 to 2021 against the global OC-CCI dataset, finding that high solar zenith angles (>70°) severely degrade image quality. Under these conditions, the longer atmospheric path and increased scattering cause standard near-infrared atmospheric correction to underestimate R_{rs} by up to 50% (-38.59% bias). They built a single-layer neural network model (NN-V) that reduced the error to -5.27% and increased the amount of usable winter satellite data nearly twenty times.

Sathyendranath et al. [7] described how the ESA Ocean Colour Climate Change Initiative (OC-CCI) creates consistent long-term ocean datasets by merging data from SeaWiFS, MODIS, MERIS, and VIIRS. They applied QAA-based band-shifting to align wavelengths across different sensors, corrected sensor biases over time, and used fuzzy logic water classification to compute per-pixel uncertainty values based on real field measurements.

Castant et al. [10] constructed a global 4D dataset of Photosynthetically Active Radiation (PAR) and diffuse attenuation (K_dPAR) from 1958 to 2022 at 0.25° resolution. By combining GlobColour satellite data with the Environmental String Model and ERA5 atmospheric reanalysis using DINEOF interpolation, they proved that weather model data can be used to accurately estimate underwater light and euphotic depth (Z_{eu}) even for historical periods.

Kulk et al. [19] used 20 years of OC-CCI satellite observations to assess global marine primary production. Using photosynthesis-irradiance models calibrated with field ^{14}C experiment data, they mapped production across various ocean provinces, showing how large-scale climate variations directly influence marine productivity over time.

D. Statistical Trend Detection and Climate Time of Emergence

Hammond et al. [8] developed a spatio-temporal Bayesian model to analyze regional chlorophyll trends and uncertainties using 20 years of OC-CCI satellite data. Using CMIP5 climate models as Bayesian priors and accounting for spatial distance and AR(1) temporal autocorrelation, they found that while the global average trend was near zero (0.08 ± 0.35% per year), 18 out of 23 ocean regions showed significant regional trends (±1.8% per year), proving that global averages hide important local changes.

E. Synthesis of Literature and Identification of Research Gaps

Reviewing these twenty foundational studies reveals four main technical gaps in existing satellite ocean color workflows:

TABLE I : THEMATIC LITERATURE FOUNDATION

Thematic Pillar	Key Citations	Focal Contribution to Pipeline
A. Arabian Sea Bloom Dynamics & Drivers	Pandi et al. [1], Shi & Wang [2], Kuttippurath et al. [3], Goes et al. [11], Lakshmi et al. [12], Asha Devi et al. [13], Raj et al. [14], Xiang et al. [15], Sarma et al. [16], Kuttippurath et al. [18]	Quantifies NRR band ratios, MLD deepening (70–80 m), vertical velocities (~1 m/day), Sverdrup critical depth shoaling, trophic disruption, hypoxia (<1.5 mg/L), and dust/storm fertilization.
B. Bio-Optical Inversion & Discrimination	Cael et al. [5], Dutkiewicz et al. [6], Bisson et al. [9], Loisel et al. [20], Baliarsingh et al. [17]	Provides multivariate remote sensing reflectance framework, dynamic γ calculation in QAA, LS2 semi-analytical inversion ($R^2 > 0.96$), and optical discrimination of green Noctiluca via 531 nm reflectance peak.
C. Radiometric Correction & Datasets	Li et al. [4], Sathyendranath et al. [7], Castant et al. [10], Kulk et al. [19]	Formulates NN-V neural atmospheric correction under high solar zenith angles (SZA > 70°), OC-CCI cross-sensor band-shifting, and 4D PAR/euphotic depth reanalysis interpolation.
D. Statistical Trend Modeling	Hammond et al. [8], Cael et al. [5], Dutkiewicz et al. [6]	Deploys Cochrane-Orcutt AR(1) autoregression, GLS covariance modeling, and multivariate SNR ≥ 2.0 criteria to isolate secular climate signals from red noise decades ahead of scalar chlorophyll.

1. Bio-Optical Spectral Misidentification: Standard operational algorithms [2], [7] rely on simple blue-green ratios that cannot distinguish mixotrophic dinoflagellates from regular diatoms, causing false-positive biomass detections during bloom events [1], [17].

2. Surface Optical and Hydrodynamic Decoupling: Most satellite studies [1], [5], [17] treat imagery as flat 2D surface layers, without connecting surface features to vertical mixing velocities, mixed-layer deepening, and thermocline shoaling available from ocean reanalysis models [2], [10], [12].

3. High Data Loss in Winter: Winter bloom monitoring in the northern basin loses substantial data when solar zenith angles exceed 70°, as standard atmospheric correction algorithms fail under long atmospheric paths and dust loads, discarding nearly half of the images [3], [4].

4. Statistical Masking from Univariate Autocorrelation: Long-term trend analyses [3], [8] often rely on scalar chlorophyll-*a* time series, where natural year-to-year swings and red-noise autocorrelation hide genuine climate trends that multi-band reflectance can reveal decades earlier [5], [6].

III. PROPOSED METHODOLOGY

Our proposed computational methodology combines satellite image processing, radiative transfer inversion, neural network atmospheric correction, and multivariate time-series statistics into a complete pipeline. The workflow consists of four sequential stages, as shown in Fig. 1.

A. Stage 1: Data Ingestion and Spatial Grid Harmonization

The study area covers the northern Arabian Sea between 10°N and 25°N latitude and 50°E and 75°E longitude [1]. Top-of-atmosphere and surface radiances are downloaded from Level-2 archives of MODIS-Aqua, VIIRS (SNPP and NOAA-20), and Sentinel-3 OLCI [4], [7]. At the same time, daily physical oceanographic and meteorological variables are obtained from the Copernicus Marine Service (CMEMS GLOBAL_MULTIYEAR_PHY_001_030) and ECMWF ERA5 reanalyses at 0.25° resolution. These include sea surface temperature (SST), mixed-layer depth (MLD), vertical velocity (w at 70 m depth), wind stress, and surface photosynthetically active radiation (PAR) [2], [10].

Data harmonization across different satellites and models is implemented in Python using libraries such as xarray, numpy, and netCDF4. Orbital swaths are reprojected onto a standardized 4 km × 4 km sinusoidal equal-area grid using bilinear interpolation [7]. Quality flags are checked to remove cloudy and corrupted pixels, including land masking (LAND), cloud and ice flags (CLDICE), high solar zenith angles (HISOLZEN), stray light (STRAYLIGHT), and atmospheric correction failures (ATMFAIL) [4], [9].

B. Stage 2: Radiometric Preprocessing and High-SZA Correction

During winter in the northern Arabian Sea, low sun angles frequently result in solar zenith angles greater than 70°, causing standard atmospheric correction methods to fail. Following the approach of Li et al. [4], we implement a Multilayer Perceptron neural network (NN-V) to process top-of-atmosphere Rayleigh-corrected radiances ($L_{rc}(\lambda)$). The network input features include five visible bands ($L_{rc}(412)$, $L_{rc}(443)$, $L_{rc}(486)$, $L_{rc}(551)$, $L_{rc}(671)$) along with the sun and sensor viewing angles: solar zenith angle (θ_s), satellite viewing zenith angle (θ_v), and relative azimuth angle ($\Delta\varphi$):

$$\vec{X} = [L_{rc}(412), L_{rc}(443), L_{rc}(486), L_{rc}(551), L_{rc}(671), \quad (1)$$

where X is the neural network input feature vector; $L_{rc}(\lambda)$ represents the Rayleigh-corrected top-of-atmosphere radiance across five visible bands (412, 443, 486, 551, and 671 nm); θ_s is the solar zenith angle; θ_v is the satellite viewing zenith angle; and $\Delta\varphi$ is the relative azimuth angle between the sun and sensor.

The neural network uses a hidden layer with 64 neurons and rectified linear unit (ReLU) activation functions. It is trained on clear-sky matchups to predict normalized water-leaving reflectance ($R_{rs}(\lambda)$) [4]:

$$R_{rs}(\lambda) = f_{\text{NN-V}}(\vec{X}) \quad (2)$$

where $R_{rs}(\lambda)$ is the water-leaving remote sensing reflectance at wavelength λ (in sr^{-1}); $f_{NN\text{-}V}$ is the function learned by the neural network; and X is the combined input vector of radiances and angles.

This neural approach removes atmospheric path scattering, recovering valid R_{rs} values in winter coastal regions and reducing retrieval bias from -38.59% down to -5.27% [4].

C. Stage 3: Bio-Optical Inversion and Narrow-Band Feature Engineering

After atmospheric correction, the reflectance spectra are inverted into inherent optical properties using the Quasi-Analytical Algorithm (QAA v6) [7], [20]. First, above-water remote sensing reflectance (R_{rs}) is converted to below-surface reflectance (r_{rs}) [9]:

$$r_{rs}(\lambda) = \frac{R_{rs}(\lambda)}{0.52 + 1.7R_{rs}(\lambda)} \tag{3}$$

where $r_{rs}(\lambda)$ is the below-surface reflectance just beneath the water surface (sr^{-1}); $R_{rs}(\lambda)$ is above-water reflectance; and 0.52 and 1.7 are standard empirical air-water transmission constants.

The ratio of backscattering to total absorption and backscattering, denoted as $u(\lambda)$, is computed analytically as:

$$u(\lambda) = \frac{-g_0 + \sqrt{(g_0)^2 + 4g_1 r_{rs}(\lambda)}}{2g_1} \tag{4}$$

where $u(\lambda)$ is the optical ratio defined as b_{bp} / (a + b_{bp}); g_0 = 0.0895 and g_1 = 0.1247 are phase-function constants; and $r_{rs}(\lambda)$ is the subsurface reflectance.

The particulate backscattering coefficient ($b_{bp}(\lambda)$) and non-water absorption ($a_{nw}(\lambda)$) are then calculated using reference wavelengths at λ_0 = 555 nm or 670 nm [9], [20].

To target green *Noctiluca scintillans* specifically, we calculate two diagnostic optical features based on the experimental findings of Pandi et al. [1] and Baliarsingh et al. [17]:

1. Noctiluca Reflectance Ratio (NRR):

$$NRR = \frac{R_{rs}(531)}{R_{rs}(510)} \tag{5}$$

where NRR is the dimensionless Noctiluca Reflectance Ratio; $R_{rs}(531)$ measures the endosymbiont backscattering peak; and $R_{rs}(510)$ captures carotenoid absorption [1].

2. Empirical Cell Density Formulation:

$$N_{cells} = 89{,}360 - 72{,}133 \times \left(\frac{R_{rs}(510)}{R_{rs}(531)}\right) \tag{6}$$

where N_{cells} is the estimated green Noctiluca cell density (in cells L^{-1}); 89,360 and 72,133 are calibration coefficients; and $R_{rs}(510)$ / $R_{rs}(531)$ is the inverse band ratio, verified for cell counts above 6,800 cells L^{-1} in the northern Arabian Sea [1].

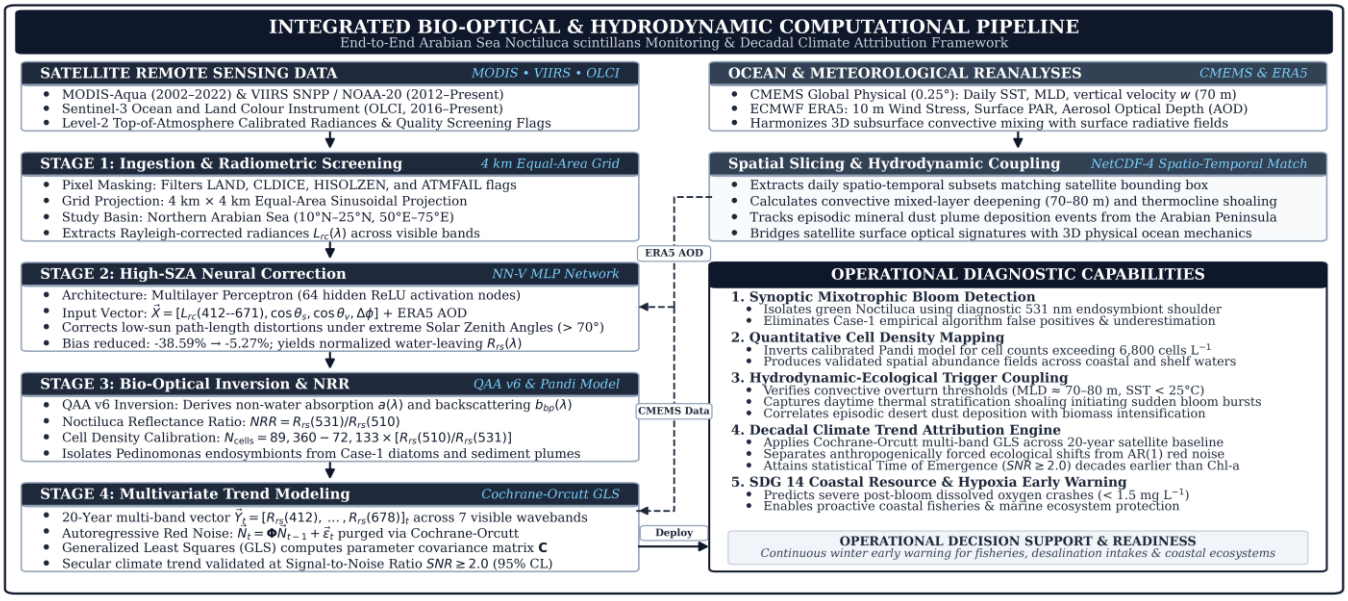


Fig. 1. End-to-end architecture of the integrated bio-optical and multivariate reanalysis pipeline.

D. Stage 4: Multivariate Autoregressive Trend Modeling

To check whether recent bloom occurrences represent real long-term climate trends or just normal decadal variations, we apply multivariate linear regression to the 20-year satellite dataset following Cael et al. [5] and Hammond et al. [8].

For every grid cell, the annual median reflectance vector across seven visible bands (Y_t = [$R_{rs}(412)$, ..., $R_{rs}(678)$]$_t$) is modeled as a linear function of time t [5]:

$$\vec{Y}_t = \vec{\mu} + \vec{\beta}t + \vec{N}_t \tag{7}$$

where Y_t is the annual median remote sensing reflectance vector across the seven visible bands in year t; μ is the baseline intercept vector; β is the trend slope vector (sr^{-1} yr^{-1}); t is time in years; and N_t is the noise vector representing environmental variations [5].

In ocean time series, the noise vector N_t contains first-order autoregressive red noise (AR(1)), meaning each year's error is correlated with the previous year:

$$\vec{N}_t = \mathbf{\Phi}\vec{N}_{t-1} + \vec{\epsilon}_t \tag{8}$$

where N_t is the noise vector in year t; $\mathbf{\Phi}$ is the lag-1 autocorrelation matrix; N_{t-1} is the noise vector from the previous year; and ε_t is uncorrelated Gaussian white noise [5], [8].

We use the Cochrane-Orcutt iterative method to estimate $\mathbf{\Phi}$ and transform the time series variables, removing autocorrelation:

$$\vec{Y}_t^* = \vec{Y}_t - \mathbf{\Phi}\vec{Y}_{t-1}, t^* = t - \mathbf{\Phi}t_{-1} \tag{9}$$

where Y_t^* is the transformed reflectance vector with autocorrelation removed; t^* is the transformed time variable; $\mathbf{\Phi}$ is the estimated lag-

1 autocorrelation matrix; and Y_{t-1} and t_{-1} represent the lagged reflectance vector and lagged time.

Generalized Least Squares (GLS) regression is then run on the transformed data to calculate the parameter covariance matrix **C** [8]. The multivariate Signal-to-Noise Ratio (*SNR*) is computed by projecting the trend vector onto the covariance uncertainty ellipse [5]:

$$SNR = \frac{\sqrt{\sum_i \beta_i^2}}{\sqrt{\left(\frac{\vec{\beta}}{\|\vec{\beta}\|}\right)^T \mathbf{C} \left(\frac{\vec{\beta}}{\|\vec{\beta}\|}\right)}} \tag{10}$$

where *SNR* is the multivariate Signal-to-Noise Ratio comparing the trend to background red noise; β_i is the trend slope in band i; $\|\beta\|$ is the overall magnitude of the trend vector; and **C** is the GLS covariance matrix [5].

A trend is considered statistically significant at the 95% confidence level if $SNR \geq 2.0$, providing an indicator of ecological change that emerges decades sooner than single-variable chlorophyll models [5], [6].

IV. EXPECTED ANALYTICAL FRAMEWORK AND DISCUSSION

Combining multi-source satellite radiometry with hydrodynamic reanalysis data provides valuable diagnostic results across three main areas:

A. Discrimination Accuracy of Mixotrophic Dinoflagellates

Using the narrow-band ratio $R_{rs}(531)/R_{rs}(510)$ allows the system to solve the common problem of confusing green *Noctiluca* with regular diatoms in standard satellite products [1], [17]. In typical diatom blooms, reflectance decreases steadily from 490 nm to 555 nm because of normal chlorophyll-*a* absorption [7]. In green *Noctiluca*, however, the green *Pedinomonas noctilucae* endosymbionts create a distinct reflectance peak near 531 nm and strong absorption near 510 nm. Tracking this specific ratio across winter satellite passes allows continuous mapping of cell densities in both coastal and offshore areas. In addition, computing dynamic particulate backscattering (b_{bp}) using the QAA model helps filter out false detections caused by mineral sediment runoff in shallow bays, ensuring that detected anomalies are real algal blooms.

B. Coupling Surface Optics with Hydrodynamic Mixing Drivers

Comparing satellite bloom masks with CMEMS physical reanalysis grids confirms the oceanographic triggers identified by Shi and Wang [2] and Lakshmi et al. [12]. During the winter northeast monsoon, areas with cooler sea surface temperatures ($SST < 25$ °C) and deeper mixed layers ($MLD \approx 70$–80 m) show significant nutrient entrainment, which is followed by bloom growth once daytime sunlight warms the surface and makes the mixed layer shallower than Sverdrup's critical depth. Furthermore, combining satellite rasters with ERA5 aerosol optical depth (AOD) data tracks desert dust storms from the Arabian Peninsula, supporting the dust fertilization mechanism described by Kuttippurath et al. [3], [18]. This shows that while winter convective mixing supplies major nutrients like nitrate and phosphate, desert dust provides essential trace metals such as iron that sustain blooms in open waters.

C. Emergence of Long-Term Ecological Trends

Using Cochrane-Orcutt multivariate regression across multi-band reflectance helps determine whether increased greening in the Arabian Sea is a permanent climate trend or part of natural decadal cycles [11]. In agreement with Cael et al. [5] and Dutkiewicz et al. [6], multivariate R_{rs} vectors reduce natural year-to-year noise, allowing significant trends ($SNR \geq 2.0$) to be detected within the available 20-year satellite record, whereas single-variable chlorophyll models remain uncertain [8]. By tracking multi-wavelength reflectance patterns, our system can detect subtle changes in plankton optical characteristics—such as rising endosymbiont carotenoids—much earlier than traditional chlorophyll indexes, providing an early warning signal for regional ecosystem shifts.

V. CONCLUSION

In this study, we presented an integrated computational Earth observation framework that connects satellite bio-optical inversion with physical hydrodynamic reanalysis to detect and monitor *Noctiluca scintillans* blooms in the northern Arabian Sea. By synthesizing twenty key papers across marine optics, radiative transfer, atmospheric correction, and time-series statistics, the proposed pipeline directly addresses four major challenges: empirical algorithm misclassification, the gap between surface satellite data and subsurface mixing, high data loss under large solar zenith angles in winter, and statistical noise in single-variable time series. Our pipeline applies diagnostic narrow-band ratios ($R_{rs}(531)/R_{rs}(510)$) to estimate cell density, uses neural network atmospheric correction to recover low-sun winter images, connects surface blooms with convective mixing reanalysis, and applies multivariate autoregressive regression to detect long-term decadal shifts. In future work, we plan to incorporate Physics-Informed Neural Networks (PINNs) into the pipeline to model 3D subsurface oxygen depletion and predict coastal hypoxia before blooms collapse, providing a practical early warning system to support marine conservation under UN Sustainable Development Goal 14.

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